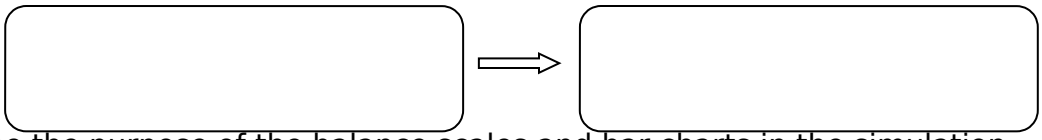
Name_____

Balancing Chemical Equations

1. Use the simulation to adjust the coefficients and balance the following equation. Select the balance scales and bar charts to help. Fill in the proper coefficients when you are successful (yellow smiley face).

____ N_2 + ____ H_2 → ____ NH_3

2. Draw the particle view of the balanced equation as shown in the simulation.



3. Describe the purpose of the balance scales and bar charts in the simulation.

4. In order for a chemical equation to be properly balanced, what must be true?

5. Balance the other two examples.

Separate of water

___ H_2O → ___ H_2 + ___ O_2

Combust of methane

___ CH_4 + ___ O_2 → ___ CO_2 + ___ H_2O

6. The number placed in front of a formula is called a coefficient. The small number within a chemical formula is called a subscript. Why do we adjust coefficients when balancing chemical equations and not subscripts?

GAME

Record your score for each of the levels in the balancing game:

	Score
Level 1	/ 10
Level 2	/10
Level 3	/ 10